One Price for a Healthy Year: Funding Bid Kit

Expression of Interest drafts, policy brief and simulated panel review

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How to use this kit

The kit contains five components:

1. **Funding landscape.** What changed at the Marsden Fund and how this project fits.
2. **Marsden-style Expression of Interest.** Summary, benefit, Vision Mātauranga and researcher statements.
3. **Health Research Council variant.** The same science framed for a health funder.
4. **Two-page policy brief.** For Treasury, the Ministry of Transport and Health New Zealand.
5. **Simulated panel review.** Three hostile referees and the responses, already built into the drafts above.

   Verify before submitting. This session could not open the Royal Society Te Apārangi guidelines. Every word or page limit below is a working target modelled on recent Marsden rounds. Check each against the current call document, and the 2027 transition-fund rules in particular.

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1 Funding landscape (as of November 2026)

- **Marsden is in transition.** In December 2024 the government refocused the Marsden Fund on "core sciences". It disbanded the social sciences and humanities panels, replaced "scholarly impact" with "quantifiable impact" and added a "benefit" section. About half of grants are expected to show economic impact.
- **The economics panel survives.** The **Economics and Human and Behavioural Sciences** panel covers "research related to human behaviour and economic systems".
- **2026 was the last traditional round.** It had about $55.8m for new grants, Fast-Start and Standard mechanisms, and no Council Awards. Funding decisions now pass to the new **Research Funding New Zealand** board, with a transition fund planned for 2027.

Strategic fit. The project is economics with a hard, quantifiable benefit: better allocation of public spending on health and safety across government. It belongs naturally on the economics panel. The

benefit case can be stated in dollars and QALYs, not just in scholarly terms.

Alternative and complementary funders:

- **Health Research Council.** Valuation of health is core HRC business; the variant is in Section 3.
- **Ministry of Transport or Waka Kotahi research programme.** Study 1 would directly inform the VSL review that follows NZTA Research Report 698.
- **Treasury.** The CBAx parameter review, possibly via a contract.
- **University of Otago research grants.** Pilot funding for the cognitive interviews and soft launch.

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2 Marsden-style Expression of Interest

2.1 Title

One price for a healthy year: reconciling how government values life and health

2.2 Lay summary (target ≤ 100 words)

New Zealand's transport agency values preventing one death at $12.5 million. Treasury values a year of healthy life at about $36,000. Spread over a typical lifetime, the road-safety figure is fourteen times higher. Inconsistencies like this mean that moving money between agencies could save more lives at no extra cost. We have developed theory that splits the gap into a few measurable parts. We will measure each part with a national survey and government data, then identify which differences are justified and which waste lives and money.

2.3 Research summary (target: one page, references allowed)

Aim. To derive, estimate and test a single value of a healthy year, the *Common Value of a Healthy Year* (v^*), that all government agencies could use. The project also estimates and publishes the few adjustments that should legitimately differ between agencies.

Background and novelty. Economists value mortality risk with the value of a statistical life (VSL) and health gains with a cost per QALY. For 40 years the literature has asked whether the two can be reconciled. Its answer has been "only under strong assumptions" (Bleichrodt & Quiggin 1999; Dolan & Edlin 2002). Our new theory asks a different question: *how* and *by how much* the bridge fails. It yields four results.

1. In a standard life-cycle model, the VSL equals v^* times discounted quality-adjusted life expectancy. This needs two separable conditions: a level condition and a marginal condition.
2. A violation of the marginal condition (illness changing the value of money) can only *raise* the VSL, and only by a second-order amount. This rules out a popular explanation for large gaps.
3. Context premia split into an anxiety component, which is a real welfare cost, and a misperception component, which is not. The two have different age profiles, so data can separate them.
4. The observed gap between any VSL and any QALY threshold factors exactly into six estimable terms: life-table arithmetic, net contribution, state dependence, context, measurement, and the shadow price of the health budget (Theorem 1).

Calibrated to New Zealand, discounting and quality adjustment *widen* the gap. The remaining factors must multiply to about 14.

Approach. Three linked studies.

- **Study 1: national stated-preference survey.** About 3,000 adults, with Māori and Pasifika boosts of 600 each. The same people value mortality-risk reductions in five contexts, health improvements, and health states. Scope tests and private-versus-tax payment vehicles are built in. Structural estimation identifies v^*, the context premia and the measurement inflation, and tests the theory's restrictions.
- **Study 2: sector shadow prices.** The marginal productivity of health spending, estimated using variation from the population-based funding formula. Revealed thresholds from PHARMAC funding decisions. Marginal benefit–cost ratios of funded transport projects.
- **Study 3: application.** About 40 recent NZ decisions re-ranked under the unified rule, with the health gain from reallocation estimated with full uncertainty.

Why now. NZTA revised its VSL in 2023, NICE changed its threshold in 2026, and NZ Treasury maintains CBAx. All three are being set independently.

2.4 Benefit statement (target ≤ 250 words)

Who benefits. Treasury, the Ministry of Transport, Waka Kotahi, Health New Zealand, PHARMAC and regulators that appraise safety or environmental rules. Ultimately, New Zealanders whose lives and health depend on how scarce public money is allocated.

Quantifiable benefit. Under illustrative assumptions, the framework implies that at the margin a dollar of health spending buys several times more health than a dollar of transport-safety spending appraised at the current VSL. If only $50m a year were reallocated or re-prioritised in line with the estimates, the illustrative models imply hundreds of additional QALYs a year. At Treasury's own

value, that is worth tens of millions of dollars annually, far more than the project's cost. Study 3 will replace these illustrations with estimates and confidence intervals.

Pathway. Results feed directly into the next CBAx parameter update and the NZTA Monetised Benefits and Costs Manual. Pre-briefing with Treasury and the Ministry of Transport is planned from month 6. All code, data and a reporting standard (three published numbers: v^*, endorsed context premia, sector shadow prices) will be released openly.

2.5 Vision Mātauranga (target ≤ 200 words)

The value of life and health is not culturally neutral. The project includes Māori and Pasifika boost samples so that their valuations are estimated precisely rather than averaged away. Study 1 will be co-designed with Māori researchers, with consultation through the Ngāi Tahu Research Consultation Committee, and Māori-led cognitive interviews. Data will be governed under Te Mana Raraunga principles.

A Māori-led citizens' panel will deliberate on whether context premia, such as those for identified victims or whānau impacts, should enter public appraisal. This may bring collective and intergenerational perspectives that an individual willingness-to-pay framework does not capture.

The single population value of v^* is designed so that valuations never fall for poorer or minority groups. Health and injury burdens fall disproportionately on Māori and Pasifika communities, so more efficient allocation of life-saving spending is also an equity benefit.

Draft for co-authoring. This statement should be rewritten with Māori collaborators before submission, not finalised by the principal investigator alone.

2.6 Researcher statement and track record (target ≤ 1 page; outline)

- **Principal investigator.** Austin Henderson, University of Otago. Applied health economics; benefit–cost frameworks for public health interventions (the CASCADE framework for school clean air, with probabilistic and value-of-information analysis); experimental economics (auditing incentives).
- **Needed associate investigators** (to recruit before submission):
    - a stated-preference and discrete-choice methodologist;
    - a Māori health researcher (co-PI level for Study 1 and the panels);
    - a health econometrician for Study 2;
    - an international adviser on VSL methods.
- **Outputs to list.** The theory paper (*One Price for a Healthy Year*, working paper 2026); the CASCADE manuscript; the auditing experiments.

2.7 Budget logic (Standard; confirm caps)

Item	3-year total (NZ$, indicative)
Principal investigator time (0.2 FTE)	150,000
Postdoctoral fellow (1.0 FTE, 3 years)	330,000
Research assistant (0.5 FTE, 2 years)	90,000
Survey fieldwork (n ≈ 3,000 incl. boosts, cognitive testing, soft launch)	250,000
Māori and Pasifika partnership roles and koha	80,000
Citizens' panels (×4)	100,000
Data access, computing and open-access fees	40,000
Travel and dissemination	30,000
Total	**≈ 1,070,000**

If the cap is lower, a Fast-Start-sized version would keep Study 1 and the theory, dropping Study 2 to revealed thresholds only and Study 3 to a smaller set of decisions. That version would cost about $360k, based on the most recent published Fast-Start cap; confirm the current figure.

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3 Health Research Council variant

Framing. HRC reviewers will care about health outcomes, equity and implementation. The emphasis shifts from the theory to the health-system consequences.

- **Title.** *Is health undervalued? Estimating a common value of a healthy year for New Zealand decision-making.*
- **Health relevance.** PHARMAC and Health New Zealand ration care against an implicit threshold. Other agencies value life at many times that threshold. The study measures whether the health budget's shadow price is high, which would indicate that health is underfunded relative to other life-saving spending, and by how much.
- **Equity.** The Māori and Pasifika boosts and Māori-led co-design (as in Section 2.5) are central aims, not add-ons. Analyses would report whether current thresholds disadvantage conditions that are more prevalent among Māori.
- **Outcomes.** A published NZ estimate of the value of a QALY with uncertainty; a health-system shadow price; a decision rule for PHARMAC and Health NZ appraisals; open survey data.
- **Design changes.** Lead with Studies 1 and 2. Reduce the transport components to the comparison needed for the shadow price.

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4 Policy brief (two pages)

One price for a healthy year: why New Zealand's agencies value life differently, and what to do about it

The problem. Waka Kotahi values a prevented road death at $12.5m. Treasury's CBAx values a healthy life-year at about $36,000. For a 40-year-old, the road figure equals about $516,000 per discounted healthy year: fourteen times Treasury's figure. If both cannot be right, money is being misallocated. Each dollar moved from a low-yield to a high-yield use would save more healthy life at no extra cost.

What explains the gap. New research at the University of Otago shows that the gap splits exactly into a small number of measurable parts.

Part	Meaning	Should it count?
Arithmetic	Discounting and quality adjustment of the years at stake	Already standard; it *widens* the gap
Context premium	People pay more to avoid dreaded or involuntary deaths	Only the anxiety part, if citizens endorse it
Measurement	Survey answers are insensitive to risk size, which inflates VSLs	No: correct for it
Budget shadow price	A tight health budget makes each health dollar worth more	Yes, as a signal for budget decisions

What it means. If the gap is mainly measurement, the road VSL should fall. If it is mainly the health budget's shadow price, health is underfunded relative to road safety. These have very different policy implications, and they can be told apart with data.

The proposal. A short reporting standard. Every agency reports health effects in discounted healthy years and values them with three published numbers:

1. v^* : one national value of a healthy year;
2. **context premia**: a short endorsed list, possibly all equal to one;
3. **sector shadow prices**: how much a dollar is worth in each budget.

What we ask of agencies.

- Data access for the funding-formula and PHARMAC analyses.
- A seat on the project advisory group.
- Agreement to pilot the reporting standard on two upcoming appraisals.

Contact. Austin Henderson, University of Otago.

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5 Simulated panel review

Three referees were asked to attack the proposal as hard as a real panel might. Their main criticisms, and the responses now built into the drafts, follow.

Referee A (theorist)

> *“Theorem 1 is an accounting identity. Any number can be decomposed into factors that multiply to it. What is the content?”*

Response. Correct as stated, and the proposal now says so. The content lies in three places:

- **Signs and bounds.** Each factor has one from theory: $\sigma \geq 1$ and second-order; misperception is flat in age; scope inflation follows a closed form.
- **Identification.** Each factor is separately identified by a distinct contrast in the design.
- **Testable restrictions.** Proposition 1(iii) implies age invariance and equality of mortality-based and morbidity-based values; Proposition 4 implies an age gradient in the anxiety premium.

The research summary now leads with these.

> *“Fair annuities and $r = \rho$ are heroic. Without annuities the VSL has a different form.”*

Response. Agreed. The net-contribution term N absorbs the main change, and the paper states it as a limitation. A robustness extension without annuities, with a bequest motive, will be added. It changes N but not the separation of σ, π and M.

Referee B (stated-preference empiricist)

> *“You are identifying anxiety from an age gradient. Age is correlated with everything: income, risk perception, exposure. This will not separate.”*

Response. The age gradient is one of three identification sources, not the only one. The design also includes:

- direct perception items, from which ω is estimated;
- worry frequency and intensity items, used as instruments for α;
- exposure measures and income controls.

The pre-analysis plan specifies that α is reported only if the age-gradient and worry-based estimates agree within their confidence intervals.

> *"A common baseline risk of 70 in 100,000 for bowel cancer is not credible."*

Response. Cognitive interviews will test credibility. A fallback arm using context-true baselines is costed. The structural model can accommodate different baselines because VSL is computed per unit of risk.

> *"Three thousand respondents across twenty cells is thin."*

Response. Context effects are estimated across vehicle and scope cells, about 600 per context. Power for the main contrast ($\pi = 1.25$) is 80% at about 450 per arm. Vehicle and scope main effects use about 1,500 per arm. Interactions are exploratory and labelled as such.

Referee C (policy economist)

> *"Even with perfect estimates, agencies will not adopt a common value; budgets are political."*

Response. The proposal does not require adoption of a single number. It makes the existing implicit numbers visible and comparable. The reporting standard is designed so that agencies keep their own context premia and shadow prices, but must publish them. The policy brief asks for a pilot on two appraisals, not wholesale reform.

> *"The 'quantifiable benefit' claim depends on your own illustrative numbers."*

Response. Agreed. The benefit statement now says explicitly that the figures are illustrative and that Study 3 will produce estimates with uncertainty. The cost of the project, about $1m, is small relative to annual life-saving expenditure across government, which runs to billions. The expected value of the information is therefore plausibly large; a formal VOI estimate could be added to the full proposal.

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6 Checklist before submission

- [ ] Confirm the 2027 funding mechanism, caps, panels and page limits against the official call.
- [ ] Recruit associate investigators, including a Māori co-investigator, and co-write Section 2.5.
- [ ] Obtain letters of support from Treasury (CBAx team), the Ministry of Transport and Health NZ.
- [ ] Post the theory paper as a working paper (for example in a University of Otago discussion paper series) before submission so that reviewers can read it.

- [ ] Have colleagues outside economics read the lay summary.
- [ ] Update the budget with the Otago research office’s overhead and salary rates.